

VR Breastfeeding: A Preliminary User Experience Study comparing Head and Eye Tracking Interactions

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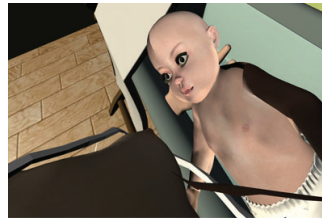
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(a) newborn being picked up for breastfeeding



(b) Newborn being positioned for latching



(c) Successful latch for breastfeeding

Figure 1: Virtual latching process.

ABSTRACT

Breastfeeding is an essential activity for newborns and mothers, with low rates becoming a major health concern. While various breastfeeding programs aim to increase the success and the desire to breastfeed using electronic health (eHealth) resources, only in-person classes provide hands-on experience with mock-up models. Such resources lack immersion and proper task representation, achievable using virtual reality (VR). This paper focuses on better understanding the perceived educational and user experience value of a VR breastfeeding latching prototype and how user interactions leveraging eye and head tracking impact usability and cognitive load.

Index Terms: Virtual reality, eye tracking, head tracking, breastfeeding.

1 INTRODUCTION

Breastfeeding is essential for the health, proper development, and survival of newborns as milk contains nutrients and antibodies. For mothers, breastfeeding reduces the risk of breast and ovarian cancer, overweight, and diabetes. Although the benefits of breastfeeding are recognized, there are ongoing efforts to increase the rate and success for the first six months to at least 50% by 2025 and 70% by 2030 in Canada [2]. In Canada, 38% of parents exclusively breastfed for the first six months, while 68.5% were doing any breastfeeding [2]. A main reason for stopping breastfeeding before 6 months included difficulty with breastfeeding (38.4%) and a lack of support or information to breastfeed [2].

Efforts to improve and increase the rate and success of breastfeeding have focused on programs that rely on videos, images, and in-person sessions as part of prenatal courses led by lactation experts. Websites and mobile applications, also known as eHealth

(technology to deliver health care services remotely), have further enabled wide access to information to complement in-person services. Freund-Azaria et al. investigated breastfeeding barriers associated with maternal motor skills, suggesting a lack of training in this domain [3]. The intersection of breastfeeding and VR has seen tools such as Virtual Feed, which uses a doll attached to a controller and hand tracking to practice breastfeeding [8], and “Real Baby - Real Family” incorporates a tangible user interface in the form of a baby with haptics and a custom visual face generator that creates a virtual face based on the photos of the parents [5].

This paper presents a VR breastfeeding experience focused on latching. The prototype features an eye tracking (ET) and head tracking (HT) version. To understand how these interactions affect the user experience, a preliminary user study focused on capturing usability [6], cognitive load [4], educational value through the Comprehensive Breastfeeding Knowledge Scale [1] and the Michigan Standard Simulation Experience Scale (MISSES) [7]. Our work is motivated by the current adoption of VR technologies as electronic health (eHealth) resources that can positively impact breastfeeding.

2 MATERIALS AND METHODS

A within-subjects study approved by Ontario Tech University’s Research Ethics Board #17150 required participants to complete latching using ET and HT. The inclusion criteria were aligned with current practices in breastfeeding teaching that welcome friends, family, or significant others regardless of their gender to increase awareness and support when breastfeeding.

2.1 Study Procedures

Participants completed the following for HT and ET: i) familiarizing themselves with the controllers and then selecting their dominant hand and skin color for the virtual mother and newborn, ii) completing a tutorial for handling the newborn, iii) performing the virtual latching as shown in Fig. 1, and iv) completing the System Usability Scale [6], the Raw NASA Task Load Index [4], the Comprehensive Breastfeeding Knowledge Scale (CBKS) [1], and the Michigan Standard Simulation Experience Scale (MISSES) [7], followed by an open question to provide feedback regarding the experience. Additionally, heart rate using a smart watch was collected

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to identify spikes in beats per minute (bpm) that could be associated with difficulty in completing tasks.

2.2 Apparatus

The development of the VR breastfeeding experience was informed by [1] and implemented for the Pico Neo 3 VR headset with support for eye tracking and the Unity game engine (2019.4.5f1). The virtual environment presents a nursery with a crib, a drawer with toys, a frame with a colorful landscape and a rainbow, and a table to practice menu interactions and newborn interactions. Virtual latching was implemented by requiring participants to properly hold the newborn against their virtual breast by observing that the chin is not tucked into its chest and the lower lip is away from the base of the nipple and outward. Because the hands are used to hold the newborn, ET and HT provide means of interacting with the content and the graphical user interface. The virtual latching provides cues to participants to know when to latch based on the infant showing signs of hunger, implemented using mouth opening and closing animations. To alleviate the physical mismatch between how a real infant and a virtual infant are held with a VR controller using grip buttons, latching was implemented using a socket, which is a game object that connects one element to another

3 PRELIMINARY RESULTS

The study involved ten participants who were evenly divided between men and women, with half having little to no VR experience and one participant having breastfed two newborns. It is important to note that this balanced gender distribution was not sought and that a statistical analysis of significance was not performed due to the small sample size. Usability (SUS) scores were acceptable above the 68 minimum score for marginal usability [6] and higher for HT ($M = 77.25$, $SD = 9.89$) than for ET ($M = 73.00$, $SD = 9.04$). The cognitive load (RTLX) was higher with HT ($M = 41.10$, $SD = 11.20$) than with ET ($M = 39.30$, $SD = 5.95$), indicating a somewhat high cognitive load [4]. The percentage of correct breastfeeding knowledge responses (CBKS) slightly favored ET ($M = 85.80$, $SD = 0.064$) over HT ($M = 83.60$, $SD = 0.062$). The educational value was higher for ET ($M = 76.00$, $SD = 0.13$) than for HT ($M = 65.00$, $SD = 0.10$), and the teaching quality was higher for ET ($M = 85.00$, $SD = 0.12$) than for HT ($M = 73.00$, $SD = 0.30$), leading to a higher overall simulation value for eye tracking ($M = 86.00$, $SD = 0.097$) than for HT ($M = 82.00$, $SD = 0.11$). It should be noted that participant 3 was the only participant who had experience breastfeeding as a mother of two, and her score remained consistent across versions without achieving a perfect score. Finally, the heart rate results indicated that the head tracking version ($M = 83.50$ bpm, $SD = 18.8$) was slightly lower compared to eye tracking ($M = 84.70$ bpm, $SD = 15.6$). Fig. 2 presents a summary of the results, and it is important to note that the results for CBKS and MISSES were normalized to be included in the same plot with SUS and RTLX.

4 DISCUSSION

This paper presented a VR breastfeeding experience focused on latching. Our prototype compares interactive modalities based on head and eye tracking to understand how these impact usability, cognitive load, fidelity, self-efficacy, educational value, and breastfeeding knowledge when performing a virtual latch while receiving information about it. Although our results are preliminary and limited in scope due to the small sample size and demographics of the participants, we believe that valuable insights and lessons learned are related to the two interactive modalities and how the participants responded to them. The head and eye tracking showed slight differences in all collected metrics. For example, head movements, which are commonly used when focusing on the content surrounding the participants, could explain the higher self-efficacy and the medium to high physical demand when using the head as a point

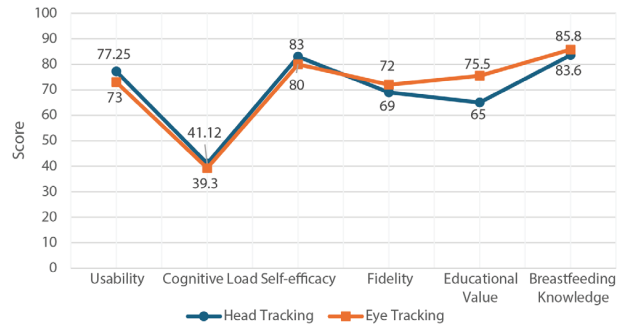


Figure 2: VR latching with head tracking and hand tracking study results.

device while also looking at the infant. Similarly, the type of tracking caused a difference in perceived fidelity, educational value, and teaching quality, where participants found the eye tracking version to be more realistic. We believe that this was the result of requiring participants to focus their attention on a specific location within the graphical user interface to navigate the content that resembles reading. Eye tracking was expected to be more usable, requiring a lower cognitive load and heart rate than head tracking. However, this was not the case, as a combination of involuntary blinking and baby manipulation issues required participants to take longer to complete latching, thus favoring the head tracking version in terms of usability. In conclusion, while preliminary, our work shows that more research is needed to better understand the role of non-hand interactions in the context of breastfeeding, as it changed how the simulation value is perceived.

Future work will focus on increasing the level of realism to better represent cues, sounds, animations, and manipulations to explore the role of embodiment and motor skill development.

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